Application of Machine Learning Models to Detect and Predict Alzheimer's Disease

Dataset: Figshare (Alzheimer's Dataset) https://figshare.com/

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1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that attacks and kills memory, thought ability, and rational capacity in such a way that the worst forms of dementia occur at the advanced stage. It impacts millions of the global population and represents a growing family and health care system strain (Bari Antor et al., 2021). Although a lot of research has been carried out, early diagnosis is a hard task due to the symptoms which usually manifest after a lot of neural damage. Old treatments like MRI and clinical observation are time-consuming, qualitative and need skilled consultancy. In this respect, large datasets can be successfully studied by machine learning (ML) methods that help to outline subtle patterns of cognitive and clinical indicators and make timely and correct predictions regarding the evolution of the disease in Alzheimer's (Kavitha et al., 2022).

1.1 Problem Definition

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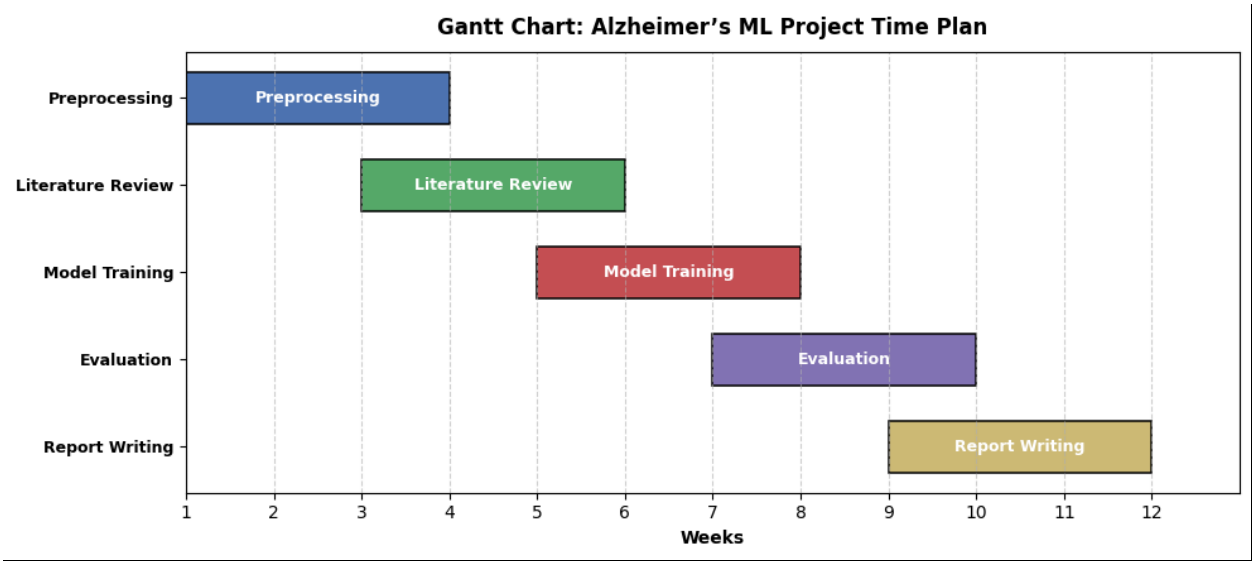

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